TASK 17 | FINAL RESEARCH REPORT

GEMSTAR Camera-LiDAR-Radar Timestamp Synchronization, Sensor Mounting Geometry and Data Validity

Project: Uncertainty-Constrained Energy-Aware ARM-FPGA Hardware-Software Co-Design for Tri-Sensor Perception

Evidence: 20261027T224313Z GT/SV OSI recording (466 frames) | Dataset: fog_mor100 folder

Status: Recorded-message alignment and mounting consistency established; acquisition-time equality and coordinate-axis conventions remain unverified.

Critical result	Observed outcome
GroundTruth / SensorView message alignment	466 / 466 identical timestamps; maximum difference 0 ms
Mounting stability	All 3 sensor configurations unchanged across 466 frames
Timing regime transition	Frame 424; 424 high-rate messages followed by 42 low-rate messages
Independent calculation	MATLAB numerical calculations agree with Python

1. Research question and motivation

Before camera-LiDAR-radar association, tracking, uncertainty estimation and hardware scheduling, are the recorded timestamps and sensor mounting configurations internally consistent? Incorrect alignment can appear as false object uncertainty or misleading processing deadlines.

2. Dataset integrity and OSI inspection | Tasks 17.1-17.2

Selected matching GT and SV files were compared against the Task 15 SHA-256 manifest. The dataset contained 466 OSI GroundTruth and 466 SensorView messages. The first SensorView contained one Camera, one LiDAR and one Radar entry, using the common SensorView timestamp; independent per-sensor capture timestamps were not established.

First recorded SensorView timestamp: 0.016666799 s; last timestamp: 28.038191422 s. Filenames and manifest entries are preserved in 05_Raw_Data_References.

3. Time extraction and reference alignment | Tasks 17.3-17.4

Python read the length-prefixed OSI messages sequentially and saved frame indices, simulated timestamps, message intervals and sensor entry counts. Every GT message timestamp was compared with the SV message of the same frame index.

Check	Result
SensorView frames	466
GroundTruth frames	466

Check	Result
Matching message timestamps	466
Maximum absolute SV-GT difference	0 ms
Median interval (all messages)	32.9931 ms
Long intervals (>2x overall median)	42

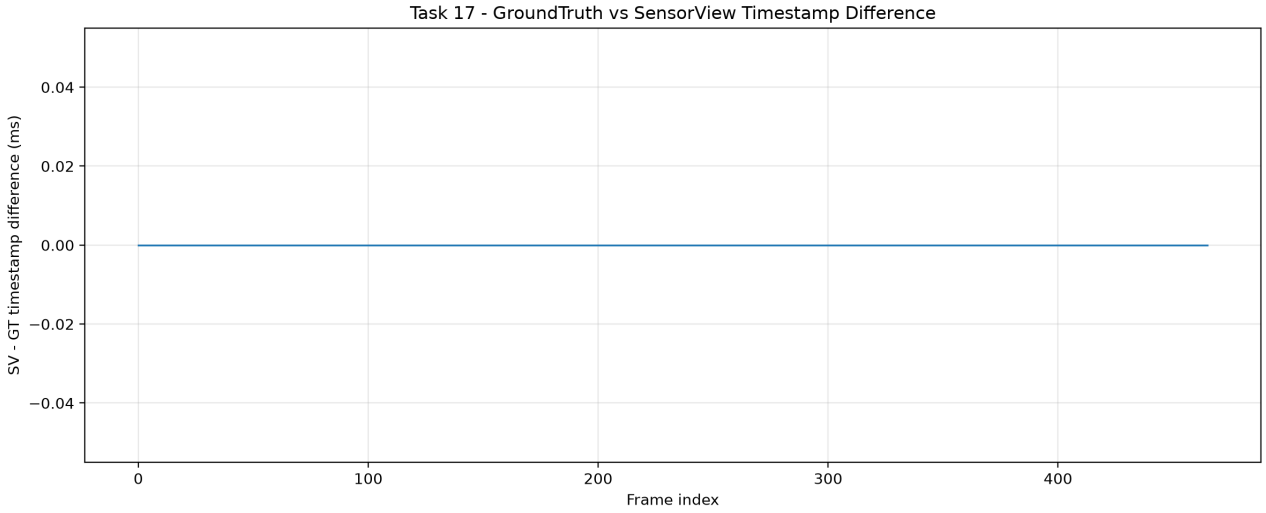


Figure 1. GroundTruth vs SensorView recorded-message timestamp difference.

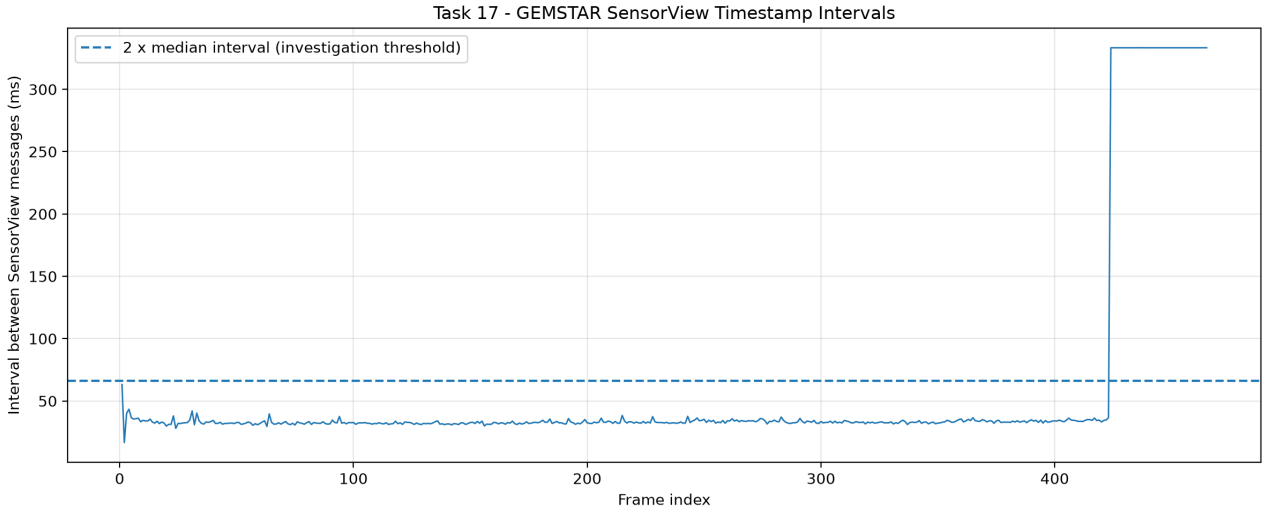


Figure 2. Interval variability across the selected OSI stream.

4. Timing regime investigation | Task 17.8

Exploratory gap analysis identified 42 long intervals, all beginning at frame 424. Recorded-message intervals changed from approximately 33 ms to approximately 333 ms. All three sensor entry types remained present in both regimes. The cause of the change was not determined; these intervals must not be interpreted as ARM or FPGA processing latency.

Segment	Frames	Median interval	Approx. message rate
HIGH_RATE	424	32.8011 ms	30.487 Hz

Segment	Frames	Median interval	Approx. message rate
LOW_RATE	42	333.3335 ms	3.000 Hz

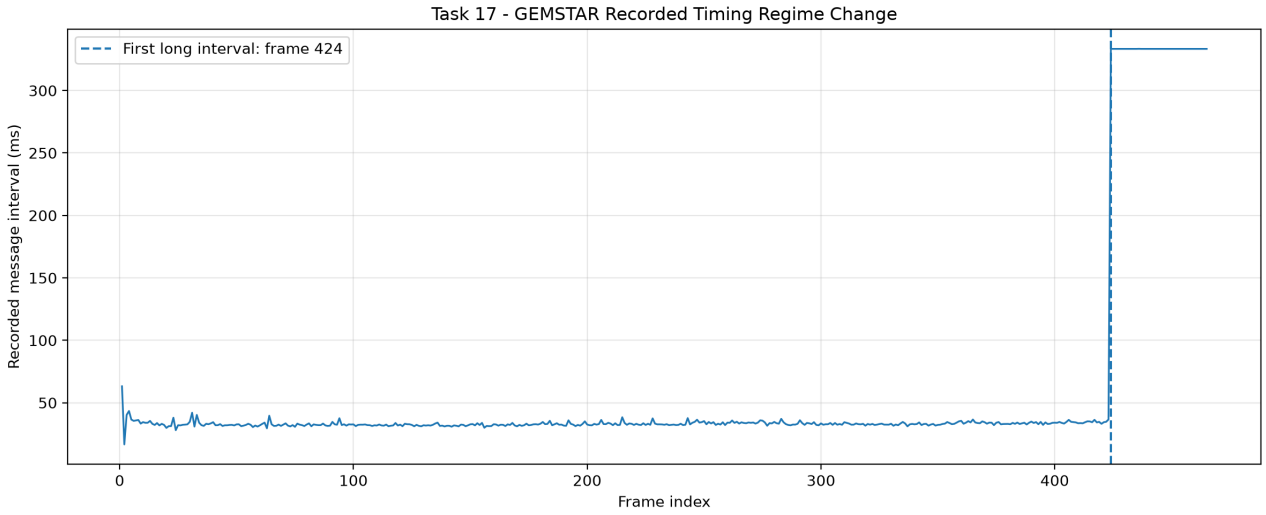


Figure 3. Distinct recorded-message timing regimes; vertical line indicates frame 424.

5. Sensor mounting geometry | Tasks 17.5-17.6

Positions extracted from OSI were stable throughout all 466 frames and numerically matched the inspected current DefaultGemstar.ini positions after cm-to-m conversion. Current configuration mounts all three sensors to SensorPlatform_1 with nominal zero Euler rotations.

Sensor	X (m)	Y (m)	Z (m)
Camera	0.700	-0.000	1.600
LiDAR	2.600	-0.000	0.700
Radar	2.000	-0.000	0.800

Inter-sensor origin distances: Camera-LiDAR 2.1024 m; Camera-Radar 1.5264 m; LiDAR-Radar 0.6083 m. Backups had different historic Radar positions and mounting assignments, so recording provenance must be distinguished from consistency with the current INI.

Three 4x4 homogeneous pose matrices were constructed from reported position and roll/pitch/yaw. Rotation orthogonality, determinant, inverse and sensor-origin mapping checks passed. These are mathematical consistency checks, not independently verified physical projection accuracy.

6. Independent MATLAB cross-validation | Task 17.7

MATLAB independently imported the timestamp and mounting CSV files, reproduced 466 exact message timestamp matches, the 32.9931 ms median interval and 42 long intervals, calculated sensor-origin distances and reconstructed three 4x4 transformations.

MATLAB metric	Value
Frames	466
ExactMatches	466

MATLAB metric	Value
MaximumErrorMs	0
MeanErrorMs	0
MedianIntervalMs	32.9931010000006
LongIntervals	42

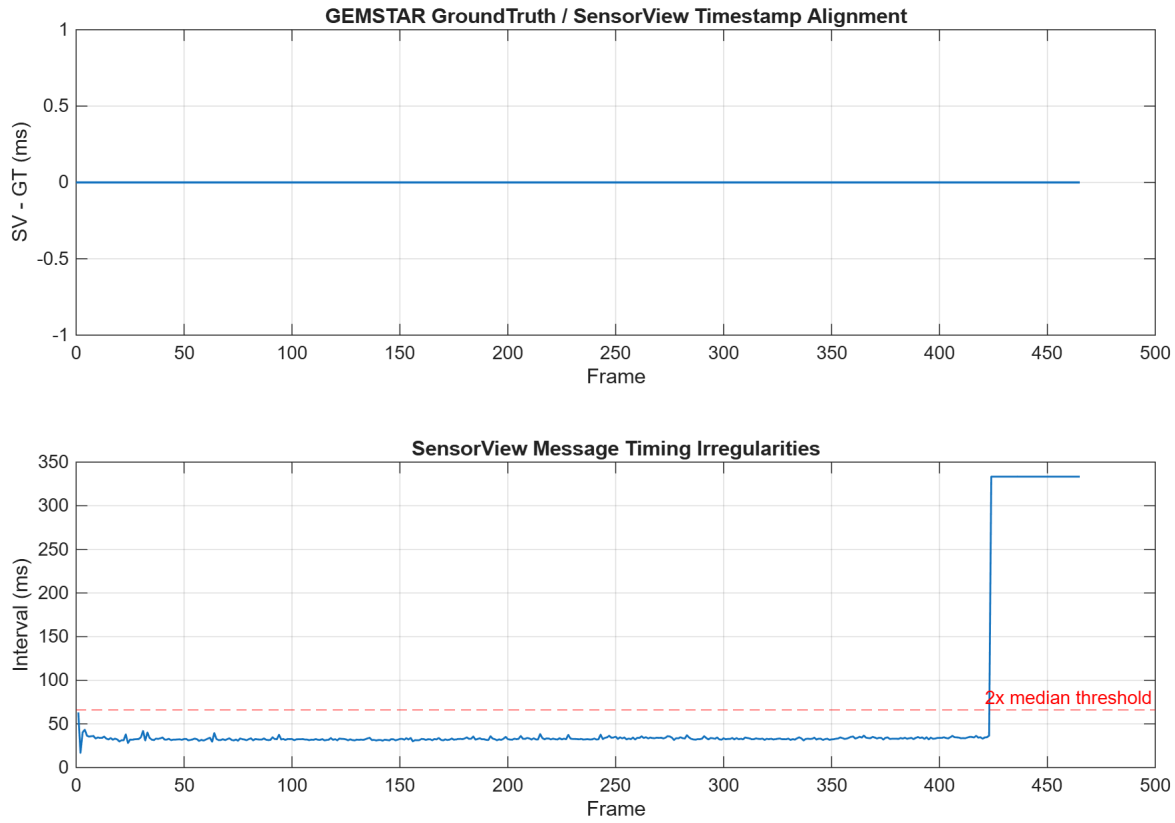


Figure 4. MATLAB timestamp alignment and message-interval visualization.

7. Source-code verification and open questions | Task 17.9

CameraSensor.cpp references ToOSI::toMountingPosition; LiDAR and Radar implementations populate mounting_position. The brief search in the main Source directory inspected only two files and found no function definition. Therefore the OSI/Unreal coordinate-axis mapping, handedness and exact geometric interpretation remain unverified. That search result does not establish that the function is absent.

8. Scientific limitations and interpretation

- Identical GT/SV timestamps establish recorded-message alignment, not independent physical sensor acquisition synchrony.

- Sensor entry presence is not a full quality or payload-correctness test.

- The reason for the 30 Hz-to-3 Hz recorded-message interval change remains unresolved.

- Sensor-to-vehicle transformations assume the reported mounting convention is correct; a known-target projection check is still needed.

- The current INI agrees with the OSI content, but its exact historical version for this recording is not proven.

9. Implications for Tasks 18-50

Tasks 18-20 will build camera image, LiDAR point-cloud and Radar preprocessing using these recording references. Tasks 21-23 must validate actual cross-sensor object projections and association against ground truth before relying on tracking uncertainty. Later hardware experiments must benchmark CPU/FPGA execution time and energy with independent timers and power instrumentation, not OSI simulated-time intervals. Separate the two recorded timing regimes during deadline experiments.

10. Evidence archive and reproducibility

01_Graphs: 4 files at report build time

02_CSV_Data: 17 files at report build time

03_Configurations: 3 files at report build time

04_Code: 12 files at report build time

05_Raw_Data_References: 1 files at report build time

06_Research_Findings: 5 files at report build time

07_Research_Report: 0 files at report build time

The original raw OSI recordings remain in the project data/raw folder; hashes and references are retained in the research archive.